IIT HYDERABAD

Department of Electrical Engineering

Electronic Devices and Circuits Laboratory Report

Experiment No: **07**



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1 Objective

- To assemble an NMOS source follower (common-drain amplifier) biased with an active current-source load constructed from a current mirror.
- To measure and characterize the DC transfer function, establishing the linear input voltage operating range and level-shift behavior.
- To determine the small-signal voltage gain (A_v) across frequency and verify near-unity performance.
- To measure the small-signal output resistance (R_{out}) using resistive loading techniques and compare it against weak-inversion theoretical predictions.

2 Components and Equipment Required

- MOSFETs : SI9926BDY dual N-channel MOSFETs on SO-8 to DIP breakout boards (3 channels used : M_{ref}, M_{cs}, M_{sf})
- Coupling Capacitor (C_{in}) 1 μF
- Resistors - 270 Ω (Reference Resistor R_{ref}) , 1 MΩ (DC Decoupling Resistor (R_{bias})) , 10 Ω, 47 Ω, 100 Ω (Load Resistors for R_{out} measurement)
- DC Power Supply
- Function Generator
- Oscilloscope (DSO)
- Digital Multimeter
- Breadboard and Jumper Wires

3 Theoretical Background

3.1 Current-Mirror Biasing

A current mirror provides an approximately constant bias current to the NMOS source follower. A reference transistor connected in a diode configuration establishes the reference current through a resistor:

$$I_{\text{ref}} = \frac{V_{\text{ref}}}{R_{\text{ref}}}. \tag{1}$$

The gate-to-source voltage of the reference transistor is applied to the current-source transistor, allowing it to mirror the reference current. This provides biasing for the source-follower stage.

3.2 NMOS Source Follower

The NMOS source follower, also known as a common-drain amplifier, has its input applied to the gate and its output taken from the source. When the transistor operates in saturation, the output voltage approximately follows the input voltage with a level shift determined by the gate-to-source voltage:

$$V_{\text{out}} \approx V_{\text{in}} - V_{GS}. \tag{2}$$

The circuit therefore provides voltage buffering with high input impedance and relatively low output impedance.

3.3 Small-Signal Voltage Gain

For small input variations around the DC operating point, the voltage gain of the source follower can be approximated as

$$A_v \approx \frac{g_m}{g_m + g_{mb} + g_{ds} + g_L}, \tag{3}$$

where g_m is the transconductance, g_{mb} is the body-effect transconductance, g_{ds} is the transistor output conductance, and g_L represents the effective conductance of the load and current-source network.

When the transistor transconductance is sufficiently large compared with the other conductance terms, the voltage gain approaches unity.

3.4 Frequency Response

The frequency response describes the variation of voltage gain with signal frequency. At low frequencies, the input coupling capacitor and the associated circuit resistance can reduce the gain. At higher frequencies, parasitic capacitances and other device effects can cause the gain to decrease.

The voltage gain in decibels is calculated as

$$A_v(\text{dB}) = 20 \log_{10} |A_v|. \tag{4}$$

The cutoff frequency is conventionally identified at the frequency where the gain falls by approximately 3 dB relative to its mid-band value.

3.5 Output Resistance

The output resistance characterizes the ability of the source follower to maintain its output voltage when a load is connected. It can be estimated experimentally by measuring the output voltage under unloaded and loaded conditions.

Using the voltage-divider model,

$$\frac{V_{\text{out,loaded}}}{V_{\text{out,unloaded}}} = \frac{R_L}{R_L + R_{\text{out}}}, \tag{5}$$

where R_L is the external load resistance. Rearranging gives

$$R_{\text{out}} = R_L \left(\frac{V_{\text{out,unloaded}}}{V_{\text{out,loaded}}} - 1 \right). \tag{6}$$

A simplified small-signal approximation for the source follower is

$$R_{\text{out}} \approx \frac{1}{g_m}, \tag{7}$$

provided other conductance contributions are negligible. The measured output resistance may differ from this approximation because of the current-source load, transistor non-idealities, and experimental conditions.

4 Mirror Verification

The current-source load is created by connecting transistor M_{ref} as a diode with reference resistor R_{ref} to set the reference current I_{ref}. The gate-to-source voltage from M_{ref} is connected to the gate of M_{cs}, mirroring the reference current to sink current from the output node.

1. Measured Values :

- Supply Voltage (V_{DD}) = 12.00 V
- Reference Resistor (R_{ref}) = 267.5 Ω
- Voltage across R_{ref} (V_{ref}) = 10.96 V (measured using Multimeter)

2. Reference Current Calculation :

Using Ohm's law across the reference resistor R_{ref}

$$\begin{aligned} I_{\text{ref}} &= \frac{V_{\text{ref}}}{R_{\text{ref}}} \\ &= \frac{10.96 \text{ V}}{267.5 \ \Omega} \\ &\approx 40.97 \text{ mA} \end{aligned}$$

The calculated current of 40.97 mA matches the expected 40 mA bias current, confirming that the current mirror operates correctly before testing the source follower.

5 DC Transfer Characteristic

5.1 Experimental Setup

The NMOS source follower (M_{sf}) is configured with its drain connected directly to the positive supply rail ($V_{DD} = 12$ V) and its source node tied to the output terminal shared with the tail current-source mirror (M_{cs}). The input gate bias V_{in} is supplied directly from an adjustable DC voltage supply (V_1), swept systematically from 1.5 V to 10.5 V in steps of 1.0 V. At each voltage step, the resulting output voltage V_{out} is measured using a Digital Multimeter (DMM) to evaluate slope linearity, level-shift behavior, and regional breakdown conditions.

5.2 Observations

Table 1: DC Transfer Characteristic (V_{out} vs. V_{in})

V_{in} (V)	V_{out} (V)	$V_{\text{in}} - V_{\text{out}}$ (V)
1.5	1.40	0.10
2.5	2.34	0.16
3.5	3.21	0.29
4.5	4.18	0.32
5.5	5.17	0.33
6.5	6.15	0.35
7.5	7.14	0.36
8.5	8.14	0.36
9.5	9.13	0.37
10.5	10.11	0.39

5.3 Graphical Analysis

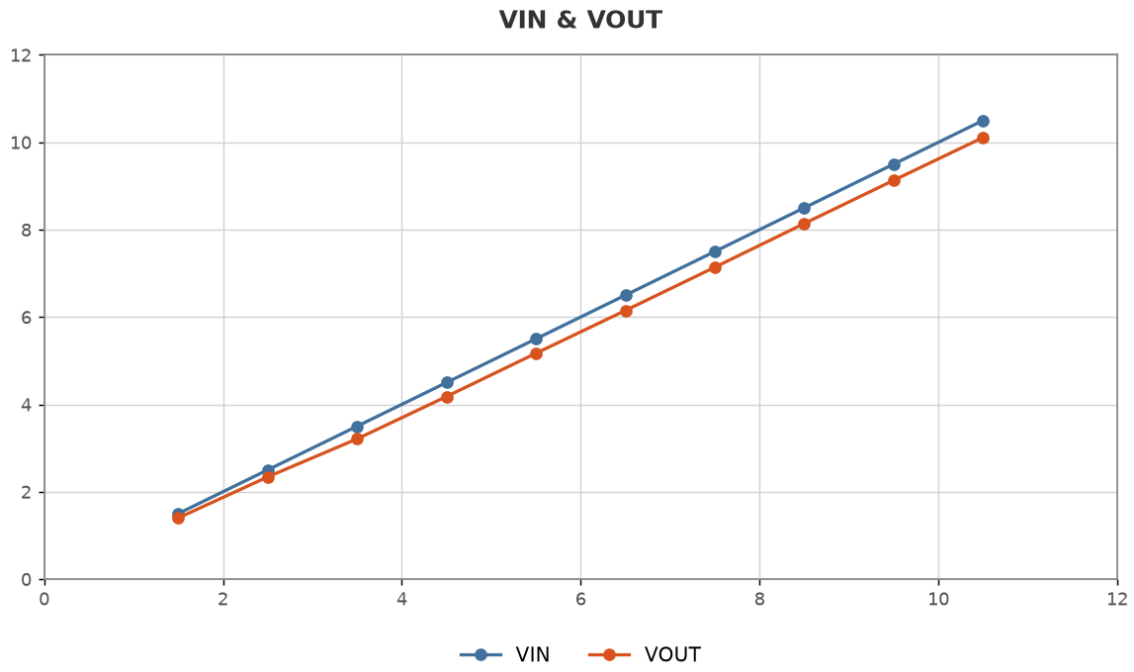


Figure 1: Experimental Plot

5.4 Simulations

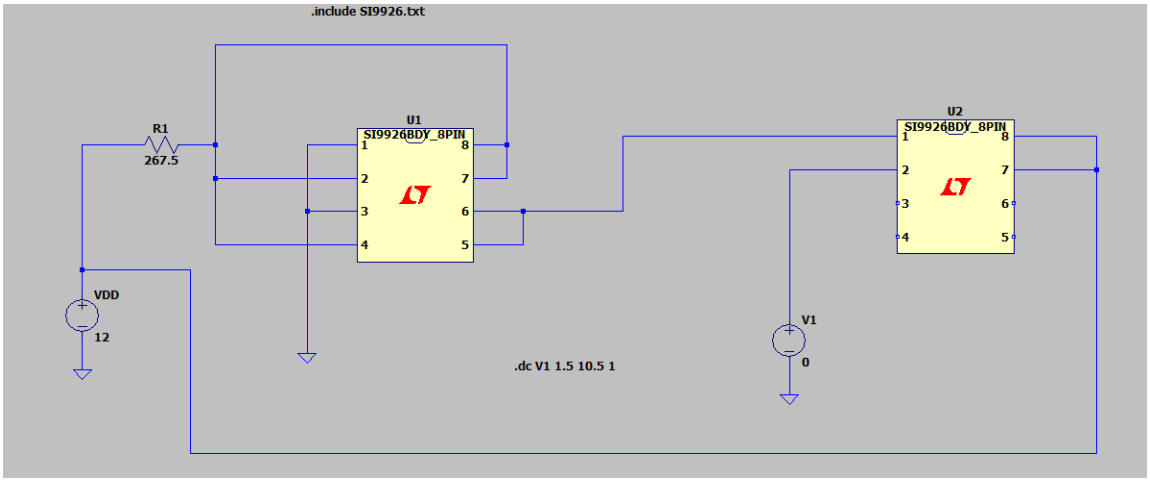


Figure 2: Circuit Schematic

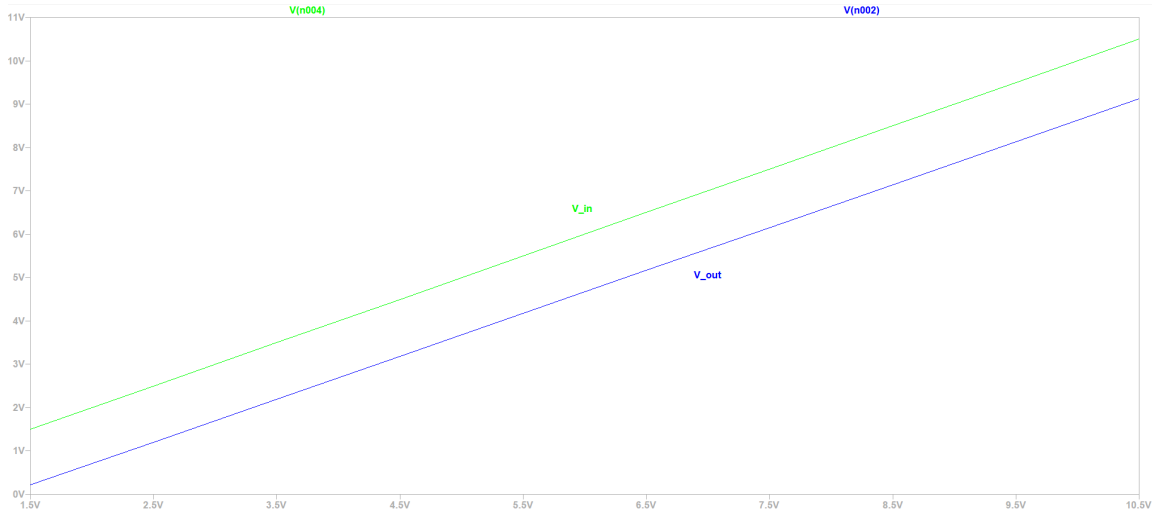


Figure 3: Simulation Plot

5.5 Calculations

1. DC Slope ($\Delta V_{\text{out}}/\Delta V_{\text{in}}$) :

Taking the linear region from $V_{\text{in}} = 3.5$ V to $V_{\text{in}} = 10.5$ V

$$\Delta V_{\text{in}} = 10.5 \text{ V} - 3.5 \text{ V} = 7.00 \text{ V}$$
$$\Delta V_{\text{out}} = 10.11 \text{ V} - 3.21 \text{ V} = 6.90 \text{ V}$$
$$\text{Slope} = \frac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} = \frac{6.90 \text{ V}}{7.00 \text{ V}} \approx 0.9857$$

The measured slope of 0.9857 is very close to the ideal slope of 1, showing accurate linear tracking across the 7 V input range.

2. Average Voltage Drop (V_{GS}) :

The relationship between input and output is given by $V_{\text{out}} = V_{\text{in}} - V_{GS}$. Calculating the average voltage difference ($V_{\text{in}} - V_{\text{out}}$) over the linear region ($V_{\text{in}} = 3.5$ V to 10.5 V) :

$$\begin{aligned} V_{\text{shift, avg}} &= \frac{0.29 + 0.32 + 0.33 + 0.35 + 0.36 + 0.36 + 0.37 + 0.39}{8} \\ &= \frac{2.77}{8} \\ &= 0.34625 \text{ V} \approx 0.35 \text{ V} \end{aligned}$$

3. Low Voltage Behavior :

At $V_{\text{in}} = 1.5$ V, V_{out} is 1.40 V, giving a much smaller difference of 0.10 V. At this low input voltage, the bottom transistor M_{cs} stops working as a constant current source and enters the triode region, breaking the linear relationship.

4. Theoretical Comparison of $V_{\text{in}} - V_{\text{out}}$ (Level-Shift) :
The relationship between input and output voltages is given by $V_{\text{out}} = V_{\text{in}} - V_{GS}$. The experimental mean voltage shift was 0.35 V, averaged over the linear saturation region ($V_{\text{in}} = 3.5$–10.5 V). This is lower than the theoretical estimate of $V_{GS} \approx 1.09$ V, calculated using $I_{\text{ref}} \approx 40.4$ mA and $V_{th} \approx 1.05$ V. The discrepancy may be attributed to device non-idealities, transistor parameter variations, and body effect.

6 Small-Signal Gain

6.1 Experimental Setup

The source follower was biased at a DC input voltage of approximately $V_{in,DC} = 7$ V. A small sinusoidal signal with an amplitude of 100 mV was applied through the coupling capacitor to the gate of the source-follower transistor. The signal frequency was set to 1 kHz.

The input and output waveforms were observed simultaneously using a Digital Storage Oscilloscope (DSO). The amplitudes of the AC components of V_{in} and V_{out} were measured and used to determine the small-signal voltage gain.

6.2 Observations from Lab

The input and output waveforms were observed simultaneously using the Oscilloscope (DSO). The output waveform closely follows the input waveform with nearly the same amplitude, demonstrating the near-unity voltage gain of the source follower.



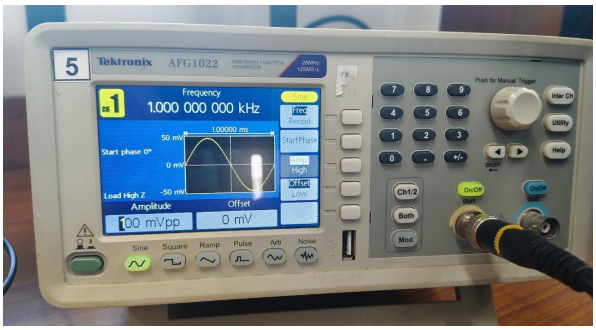


Figure 4

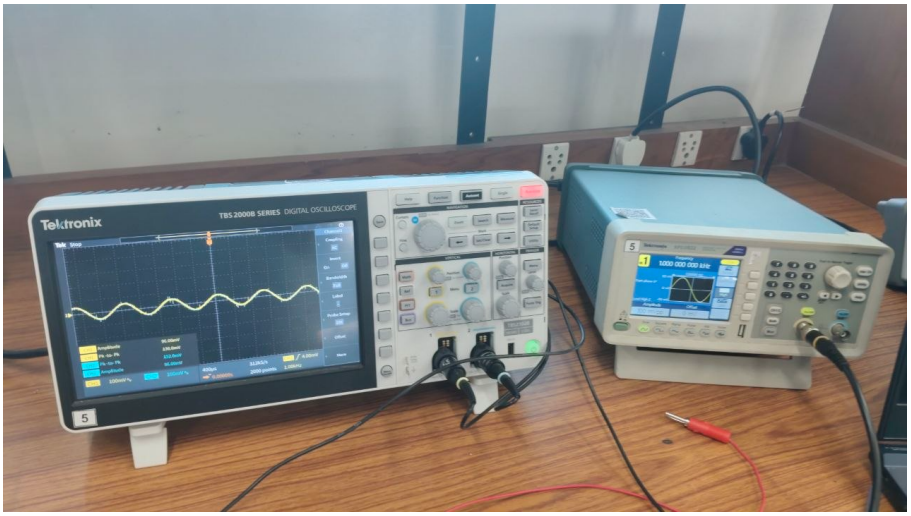


Figure 5

6.3 Observations

The measured values are shown in Table 2.

Table 2: Small-Signal Gain Measurement

Parameter	Measured Value
DC input bias, $V_{in,DC}$	7.00 V
Frequency	1 kHz
AFG amplitude	100 mV
Input AC amplitude, $V_{in(ac)}$	96.00 mV
Output AC amplitude, $V_{out(ac)}$	95.60 mV

6.4 Simulations

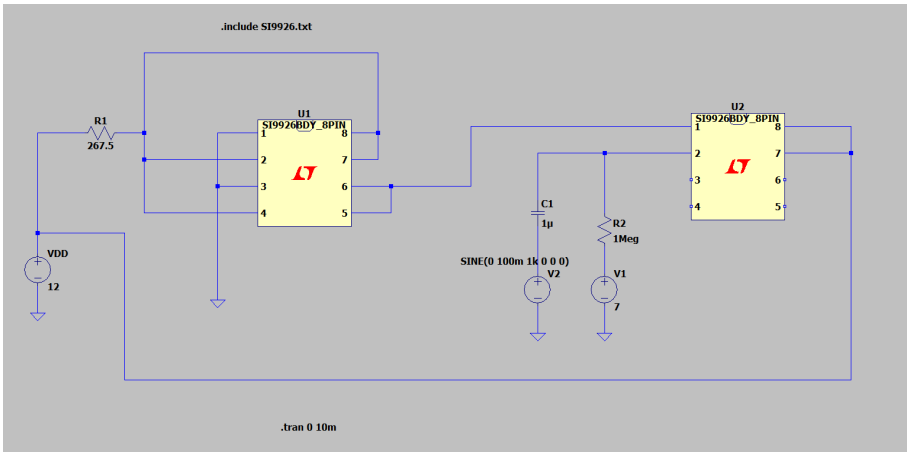


Figure 6: Schematic

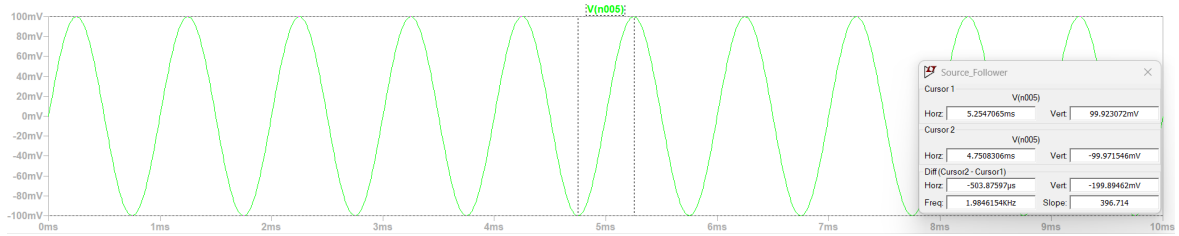


Figure 7: Small signal Input to the circuit

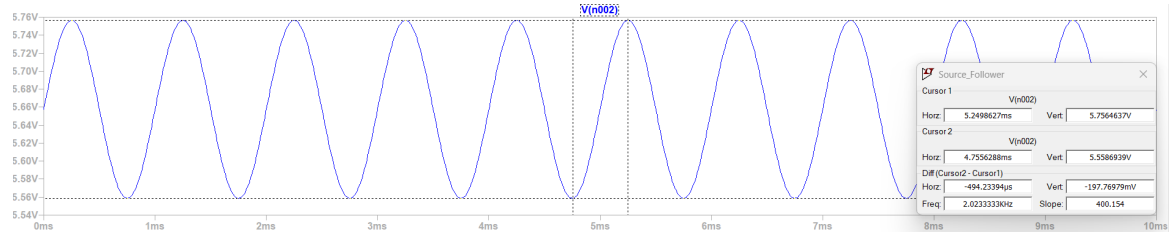


Figure 8: Output for the Input signal above

The input signal is a sinusoidal waveform with an amplitude of 100 mV and a frequency of 1 kHz. The output response is observed through simulation. From the waveform, the measured input and output amplitudes are approximately 199.89462 mV and 197.76979 mV, respectively. So the gain is calculated as

$$|A_v| = \left| \frac{197.76979 - (-197.76979)}{199.89462 - (-199.89462)} \right|$$

$$|A_v| = \frac{395.53958}{399.78924}$$

$$|A_v| \approx 0.9894$$

6.5 Calculations

The small-signal voltage gain is given by

$$A_v = \frac{V_{out(ac)}}{V_{in(ac)}}. \tag{8}$$

Using the measured values,

$$A_v = \frac{95.60 \text{ mV}}{96.00 \text{ mV}} \tag{9}$$

$$= 0.9958. \tag{10}$$

Thus, the measured small-signal voltage gain is

$$\boxed{A_v = 0.9958 \text{ V/V}}. \tag{11}$$

The measured gain is in close agreement with the theoretical near-unity gain expected from a source follower.

7 Frequency response

7.1 Experimental Setup

The frequency response of the NMOS source follower was measured using the AC-coupled configuration. A sinusoidal input signal of approximately 100 mV was applied using the function generator. The input signal was coupled to the gate through the 1 μF coupling capacitor, while the DC bias was maintained using the 1 MΩ bias resistor.

The frequency of the input signal was varied from 10 Hz to 100 kHz. At each frequency, the input and output amplitudes were measured using the digital storage oscilloscope (DSO). The input amplitude was kept approximately constant while taking the measurements.

7.2 Observations

The measured input and output amplitudes are shown in Table 3.

Table 3: Measured frequency response

Frequency (Hz)	V_{in} (mV)	V_{out} (mV)	$\lvert A_v \rvert$	Gain (dB)
10	96	76	0.792	-2.03
30	96	84	0.875	-1.16
100	92	92	1.000	0.00
300	96	92	0.958	-0.37
1000	96	95.6	0.996	-0.04
3000	96	100	1.042	0.35
10000	100	100	1.000	0.00
30000	100	100	1.000	0.00
100000	100	100	1.000	0.00

The voltage gain was calculated using

$$|A_v| = \frac{V_{\text{out}}}{V_{\text{in}}}. \tag{12}$$

The gain in decibels was calculated using

$$A_v(\text{dB}) = 20 \log_{10} |A_v|. \tag{13}$$

The gain approaches unity from approximately 100 Hz onwards. At very low frequencies, the output amplitude is lower than the input amplitude. The lowest measured gain occurs at 10 Hz, where the gain is approximately 0.792 or -2.03 dB.

7.3 Graphical Analysis

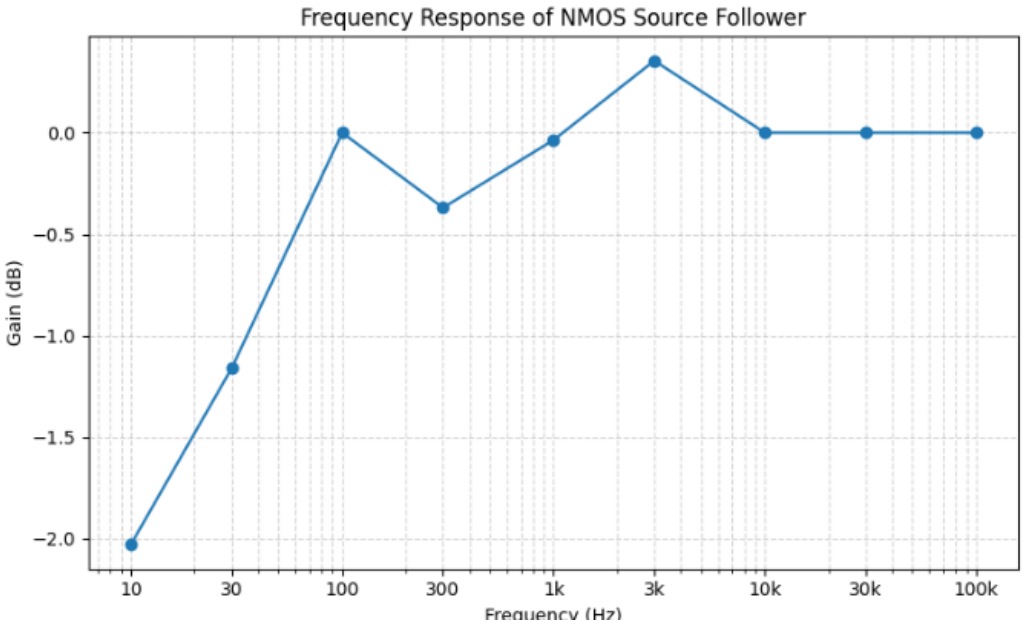


Figure 9: Experimental Plot

A plot of gain in decibels versus frequency was obtained using a logarithmic frequency axis. The measured response shows a low-frequency roll-off, with the gain decreasing to -1.16 dB at 30 Hz and -2.03 dB at 10 Hz. At approximately 100 Hz and above, the gain remains close to 0 dB, indicating a near-unity voltage gain.

At 10 Hz, the measured gain is

$$A_v = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{76}{96} = 0.792, \tag{14}$$

which corresponds to

$$A_v(\text{dB}) = 20 \log_{10}(0.792) \approx -2.03 \text{ dB}. \tag{15}$$

The reduction in gain at low frequency can be attributed to the input coupling network, particularly the 1 μF coupling capacitor and the associated input resistance. As the frequency increases, the effect of the coupling capacitor decreases and the gain approaches unity.

For example, at 1 kHz,

$$A_v = \frac{95.6}{96} = 0.996, \tag{16}$$

giving

$$A_v(\text{dB}) \approx -0.04\,\text{dB}. \tag{17}$$

This confirms that the source follower exhibits nearly unity gain in the mid-band region. No significant high-frequency roll-off is observed up to 100 kHz.

The conventional corner frequency is defined at the -3 dB point. Since the lowest measured gain is -2.03 dB at 10 Hz, the -3 dB point is not reached within the measured frequency range. Therefore, the exact corner frequency cannot be determined from the available measurements, but it can be concluded that

$$f_c < 10\,\text{Hz}. \tag{18}$$

7.4 Simulations

The frequency response of the source follower circuit was simulated in LTspice using an AC Analysis directive (`.ac`). Small-signal AC excitation was applied via voltage source V_2 with an AC amplitude of 1 V through coupling capacitor $C_1 = 1$ μF. The DC operating point was established with $V_1 = 7.0$ V through $R_2 = 1$ MΩ.

A decade frequency sweep was performed from 10 Hz to 1 GHz with 100 points per decade

```
.ac dec 100 10 1G
```

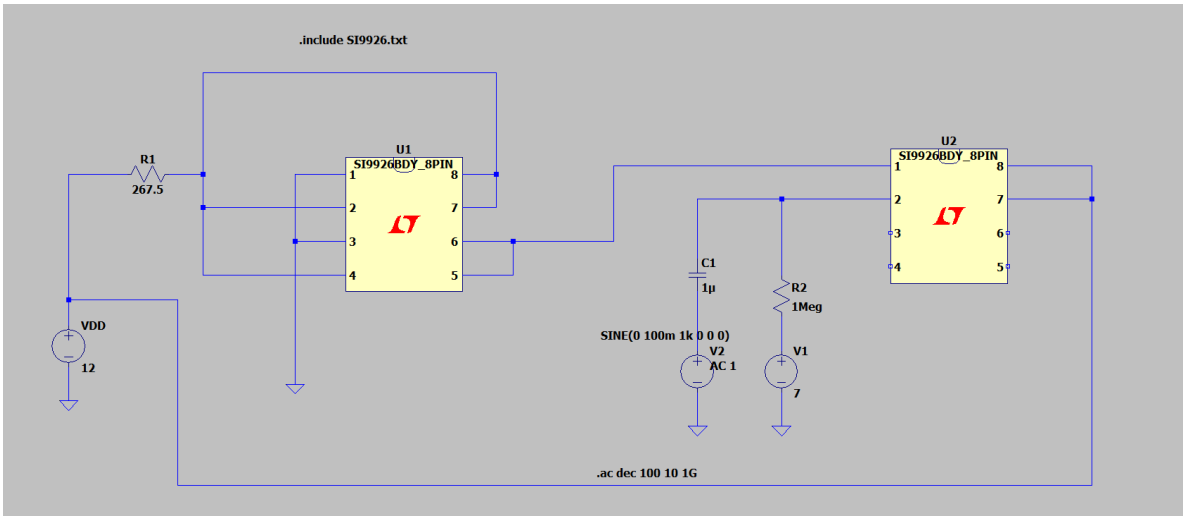


Figure 10: LTspice Schematic

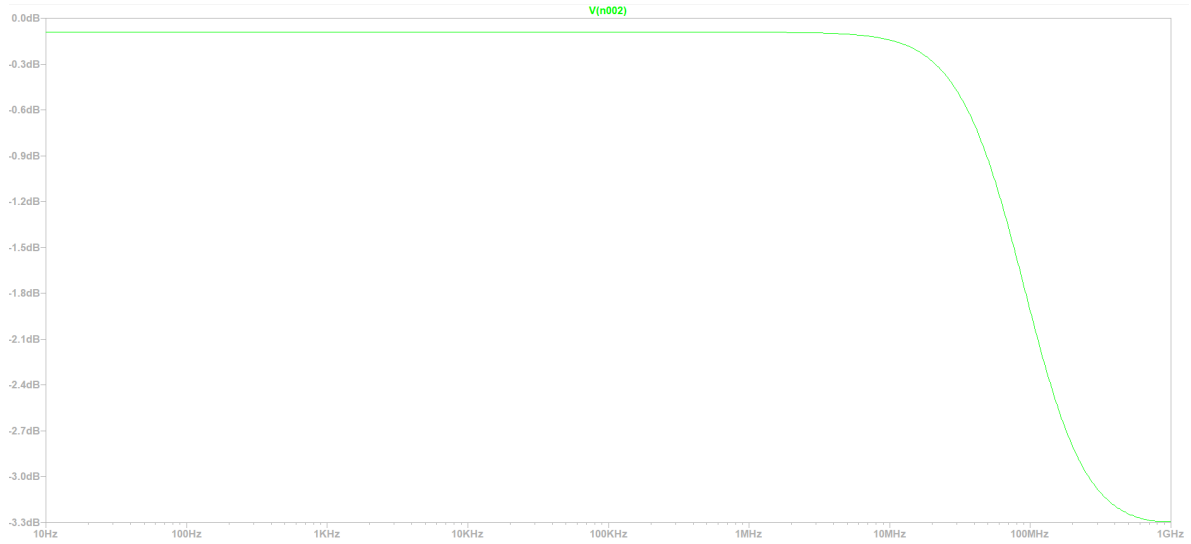


Figure 11: Simulated AC Frequency Response (V_{out} vs. Frequency)

Simulation Results:

- **Mid-Band Gain (A_v):** The passband response shows a flat mid-band voltage gain of -0.10 dB (0.9886 V/V), verifying near-unity tracking.
- **High-Frequency Cutoff (f_H):** The upper 3-dB cutoff frequency occurs at 190 MHz, where the response drops by 3 dB to -3.10 dB.
- **Low-Frequency Cutoff (f_L):** The response remains flat down to 10 Hz, confirming that $C_1 = 1\ \mu$F provides effective DC decoupling without limiting low-frequency signal transmission.

8 Output resistance

8.1 Experimental Setup

The output resistance of the NMOS source follower was determined using the loaded-output method. The circuit was operated at a frequency of 1 kHz, where the voltage gain was close to unity.

First, the output voltage was measured without any external load. The unloaded output voltage was found to be

$$V_{\text{out,unloaded}} = 95.6\,\text{mV}.$$

External load resistors of $10\,\Omega$, $48\,\Omega$, and $100.1\,\Omega$ were then connected to the output, and the corresponding loaded output voltages were measured.

The output resistance was determined using

$$\frac{V_{\text{out,loaded}}}{V_{\text{out,unloaded}}} = \frac{R_L}{R_L + R_{\text{out}}}.$$

Rearranging,

$$R_{\text{out}} = R_L \left(\frac{V_{\text{out,unloaded}}}{V_{\text{out,loaded}}} - 1 \right).$$

8.2 Observations

The measured values are shown in Table 4.

Table 4: Output resistance measurements

R_L (Ω)	V_{in} (mV)	V_{out} (mV)	$\frac{V_{\text{out}}}{V_{\text{out,unloaded}}}$	R_{out} (Ω)
10.0	94	40.0	0.418	13.9
48.0	120	68.0	0.711	19.5
100.1	116	80.0	0.837	19.5

The calculated output resistance values are approximately 13.9 Ω, 19.5 Ω, and 19.5 Ω for the 10 Ω, 48 Ω, and 100.1 Ω load conditions, respectively.

8.3 Graphical Analysis

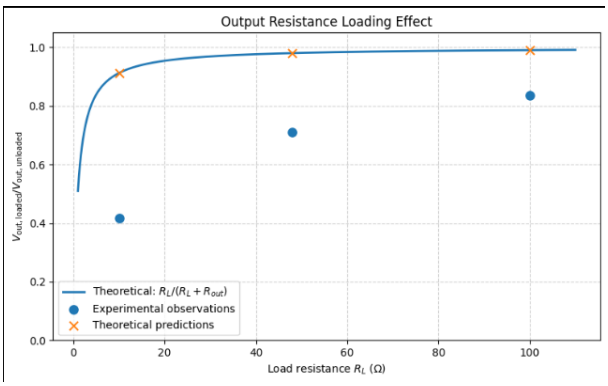


Figure 12

The objective of this analysis is to study how the load resistance affects the output voltage of the source follower and to compare the experimental results with the theoretical predictions.

When a load resistor R_L is connected to a circuit having a finite output resistance R_{out}, the output voltage decreases due to the voltage-divider effect. The ratio of the loaded output voltage to the unloaded output voltage is given by

$$\frac{V_{\text{out,loaded}}}{V_{\text{out,unloaded}}} = \frac{R_L}{R_L + R_{\text{out}}}.$$

From Section 9, the theoretical output resistance of the source follower is estimated as

$$R_{\text{out}} \approx \frac{1}{g_m} = \frac{1}{1.04} \approx 0.96\,\Omega.$$

Therefore, the theoretical voltage ratio can be expressed as

$$\boxed{\frac{V_{\text{out,loaded}}}{V_{\text{out,unloaded}}} = \frac{R_L}{R_L + 0.96}}$$

Experimental procedure: The output voltage was measured first under unloaded conditions and then with different load resistances connected to the output. The loaded-to-unloaded voltage ratio was calculated for each load resistance. These experimental ratios were plotted against R_L, along with the theoretical curve obtained using the above expression.

The experimental observations are summarized below.

$R_L\,(\Omega)$	$V_{\text{in}}\,(\text{mV})$	$V_{\text{out}}\,(\text{mV})$	Experimental ratio
10.0	94	40	0.418
48.0	120	68	0.711
100.1	116	80	0.837

Using the theoretical output resistance, the predicted ratios are

$$R_L = 10\,\Omega: \quad \frac{10}{10 + 0.96} \approx 0.912,$$

$$R_L = 48\,\Omega: \quad \frac{48}{48 + 0.96} \approx 0.980,$$

$$R_L = 100.1\,\Omega: \quad \frac{100.1}{100.1 + 0.96} \approx 0.990.$$

The comparison between the experimental and theoretical values is shown below.

$R_L\,(\Omega)$	Experimental ratio	Theoretical ratio
10.0	0.418	0.912
48.0	0.711	0.980
100.1	0.837	0.990

The graph shows that the experimental voltage ratio increases with increasing load resistance. This is the expected qualitative behaviour: a larger load resistance draws less current and causes a smaller voltage drop across the source follower's output resistance.

Consequently, the loaded output voltage approaches the unloaded output voltage as R_L increases.

However, the experimental ratios are significantly lower than the theoretical predictions. This indicates that the theoretical output resistance of $0.96\,\Omega$ does not quantitatively explain the measured loading effect. Possible reasons include measurement uncertainty, differences between the actual and estimated output resistance, and variations in the operating point of the transistor. The discrepancy should therefore be acknowledged rather than treating the experimental and theoretical values as consistent.

In conclusion, the experiment verifies the qualitative relationship between load resistance and output voltage, but the quantitative disagreement suggests that the theoretical model or the experimental measurements require further investigation.

8.4 Simulations

The output resistance R_{out} and loading behavior of the NMOS source follower were evaluated by performing a parametric sweep over load resistances $R_L \in \{10\ \Omega, 47\ \Omega, 100\ \Omega\}$ connected at the output node. The input source V_2 supplied a 1 V AC signal, and the parameter sweep was executed using the LTspice directive:

```
.step param RL list 10 47 100
```
(19)

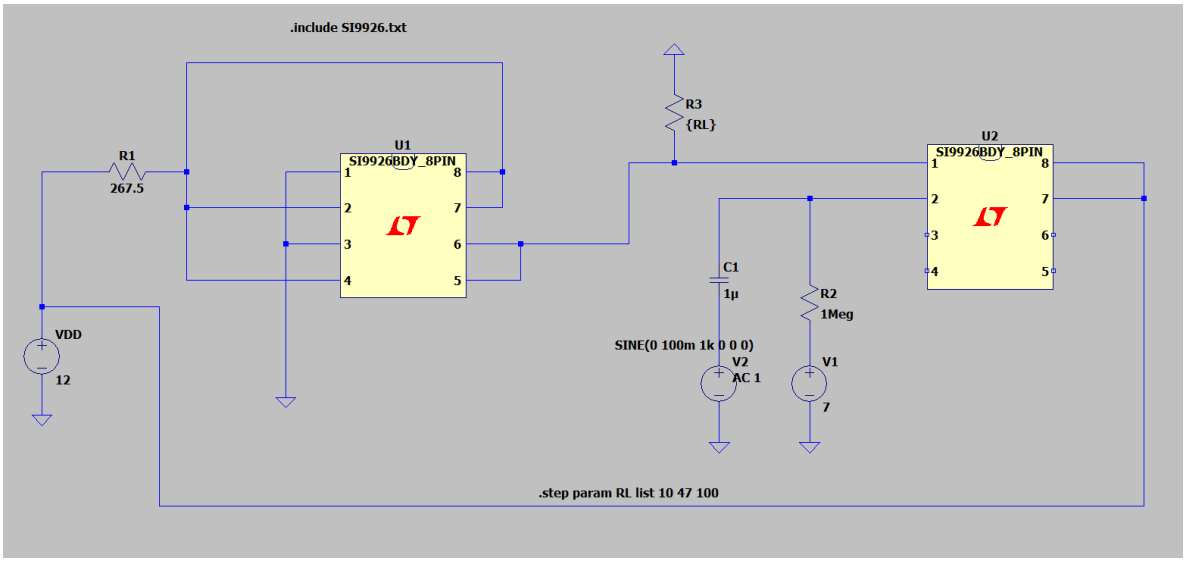


Figure 13: LTspice Schematic

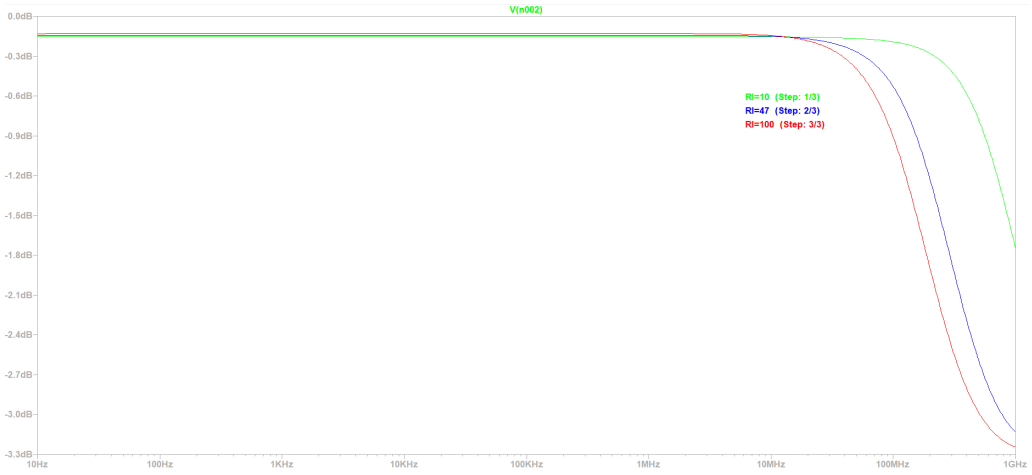


Figure 14: Simulated AC Frequency Response across Load Resistances ($R_L = 10\ \Omega, 47\ \Omega, 100\ \Omega$)

Simulation Results:

- **Loading Effect :** Decreasing the load resistance R_L from 100 Ω to 10 Ω attenuates the mid-band voltage gain due to voltage division between the circuit's output resistance R_{out} and R_L.
- **High-Frequency Bandwidth :** Smaller values of R_L lower the effective node resistance at V_{out}, shifting the high-frequency 3-dB cutoff f_H higher across the frequency spectrum.
- **Output Resistance Extraction :** The small-signal output resistance R_{out} is extracted using the open-circuit output voltage $V_{\text{out, open}}$ alongside the loaded voltage $V_{\text{out}}(R_L)$

$$R_{\text{out}} = R_L \left(\frac{V_{\text{out, open}}}{V_{\text{out}}(R_L)} - 1 \right) \tag{20}$$

8.5 Calculations

The output resistance was calculated using

$$R_{\text{out}} = R_L \left(\frac{V_{\text{out,unloaded}}}{V_{\text{out,loaded}}} - 1 \right),$$

where

$$V_{\text{out,unloaded}} = 95.6\,\text{mV}.$$

For $R_L = 10\,\Omega$ and $V_{\text{out,loaded}} = 40\,\text{mV}$,

$$R_{\text{out}} = 10\left(\frac{95.6}{40} - 1\right)$$

$$\boxed{R_{\text{out}} \approx 13.9\,\Omega}.$$

For $R_L = 48\,\Omega$ and $V_{\text{out,loaded}} = 68\,\text{mV}$,

$$R_{\text{out}} = 48\left(\frac{95.6}{68} - 1\right)$$

$$\boxed{R_{\text{out}} \approx 19.5\,\Omega}.$$

For $R_L = 100.1\,\Omega$ and $V_{\text{out,loaded}} = 80\,\text{mV}$,

$$R_{\text{out}} = 100.1\left(\frac{95.6}{80} - 1\right)$$

$$\boxed{R_{\text{out}} \approx 19.5\,\Omega}.$$

The average experimental output resistance is

$$R_{\text{out,avg}} = \frac{13.9 + 19.5 + 19.5}{3}$$

$$\boxed{R_{\text{out,avg}} \approx 17.6\,\Omega}.$$

For comparison, the theoretical value given in Section 9 is

$$g_m \approx 1.04\,\text{A/V},$$

and hence

$$R_{\text{out,theory}} \approx \frac{1}{g_m} = \frac{1}{1.04} \approx 0.96\,\Omega.$$

The predicted loading ratio is therefore

$$\frac{V_{\text{out,loaded}}}{V_{\text{out,unloaded}}} = \frac{R_L}{R_L + 0.96}.$$

For $R_L = 10\,\Omega$,

$$\frac{10}{10 + 0.96} \approx 0.912.$$

For $R_L = 47\,\Omega$,

$$\frac{47}{47 + 0.96} \approx 0.980.$$

For $R_L = 100\,\Omega$,

$$\frac{100}{100 + 0.96} \approx 0.990.$$

Thus, the theoretical model predicts a much smaller output resistance than the experimental value.

9 Discussion

The frequency-response measurements show that the NMOS source follower provides a voltage gain close to unity over most of the measured frequency range. At low frequencies, the gain decreases, with the largest reduction occurring at 10 Hz. The gain increases towards unity by approximately 100 Hz and remains close to unity up to 100 kHz.

At 1 kHz, the measured voltage gain is approximately 0.996, or -0.04 dB. This confirms the near-unity voltage gain expected from a source follower.

The output-resistance measurements were performed using load resistances of 10 Ω, 48 Ω, and 100.1 Ω. For the 10 Ω load, the measured output voltage was 40 mV, giving an output resistance of approximately 13.9 Ω. For the 48 Ω and 100.1 Ω loads, the calculated output resistances were approximately 19.47 Ω and 19.54 Ω, respectively.

The results show that the calculated output resistance is of the same order for all three load conditions, although the 10 Ω case shows a larger deviation from the other two measurements. This difference may be attributed to measurement uncertainty and the greater sensitivity of the output voltage to measurement errors when a low-value load resistance is used.

The average output resistance obtained from the three measurements is approximately

$$R_{\text{out,avg}} \approx 17.6\,\Omega.$$

The measured output resistance is considerably higher than the theoretical value of approximately 0.96 Ω obtained from the simplified relation

$$R_{\text{out}} \approx \frac{1}{g_m}.$$

The difference may arise from non-idealities of the MOSFET, the finite output resistance of the current-source load, variations in the actual bias current, and experimental measurement limitations.

Thus, the simplified theory predicts a very low output resistance, while the experimental results give a higher output resistance for the complete circuit under the actual

operating conditions.

10 Conclusion

The frequency response and output resistance of the NMOS source follower with a current-mirror load were experimentally investigated.

The frequency-response measurements showed that the voltage gain becomes approximately unity above the low-frequency region and remains nearly constant up to 100 kHz. At 10 Hz, the measured gain was approximately 0.792, while at 1 kHz the gain was approximately 0.996, confirming the near-unity gain characteristic of the source follower.

The output-resistance measurement was performed using 10 Ω, 48.0 Ω, and 100.1 Ω load resistors. The corresponding output-resistance values were approximately 13.9 Ω, 19.47 Ω, and 19.54 Ω, respectively. The average experimental output resistance was approximately 17.6 Ω.

Although the 10 Ω measurement shows some deviation from the values obtained with the larger load resistances, all three measurements indicate a finite and relatively low output resistance.

Overall, the experiment demonstrates the near-unity voltage gain and buffering behaviour of the NMOS source follower. It also demonstrates that the practical output resistance of the complete circuit can differ significantly from the simplified theoretical value of $R_{\text{out}} \approx 1/g_m$ due to the non-ideal characteristics of the circuit and experimental limitations.